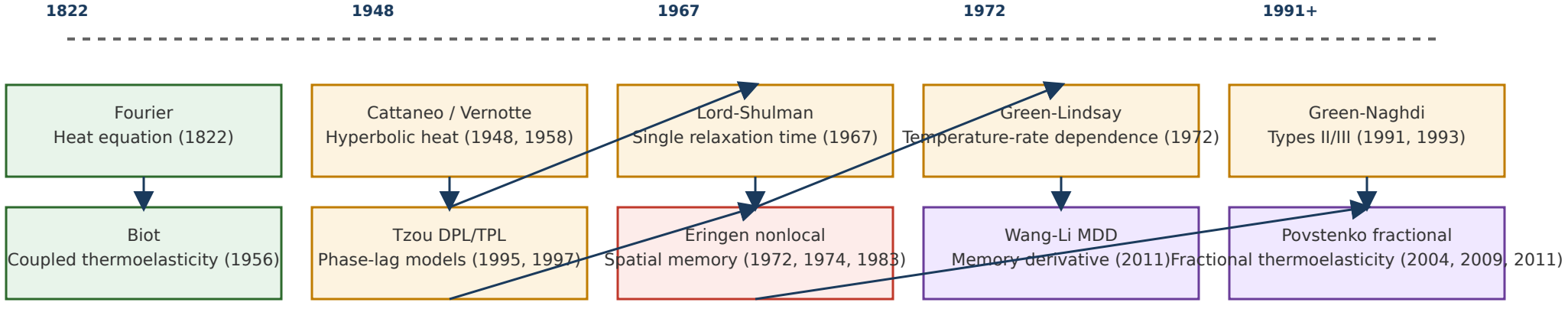
Theory family evolution in generalized continuum mechanics

Historical development of major theory families from classical to modern multiphysics formulations (Section 2)



Classical

Fourier heat conduction
Classical coupled thermoelasticity
(Biot, 1956)

Infinite thermal speed

Generalized

LS, GL, GN-II/III
DPL, TPL, MGT
Model-dependent propagation

Relaxation times

Nonlocal

Eringen nonlocal
Strain-gradient
Size-dependent waves

Spatial memory

Fractional

MDD, fractional
Thermoviscoelastic
Hybrid models

Temporal memory

Reading the figure

Arrows are schematic links; they do not assert direct derivation or equivalence.
Cross-links are provisional and require historical source checking.
Diagonal purple arrows show fractional/nonlocal connections across families.
Bottom boxes group theories by their dominant physical mechanism.
Propagation and support properties depend on the exact constitutive equations.
Source: inherited literature inventory; dates and relationships are not fully source-checked.